1.5 Solving Linear Systems

Name _____ Period ____ Date _____

Algebra II 1.5 · Student sheet

I can…

- solve a system of three linear equations by elimination.
- solve systems by substitution and recognize systems with no solution or infinitely many solutions.
- write and solve three-variable systems for real-life problems.

Vocabulary · fill in the missing word

Linear equation in three variables — ax + by + cz = d; its _____ is a plane

Solution of a system — an _____ triple (x, y, z) that makes every equation true

No solution — _____ gives a false statement, such as 0 = 5

Infinitely many solutions — _____ gives an identity, such as 0 = 0

Part 1 · Mystery numbers

Puzzle 1: Three numbers add to 20. The first is twice the second. The third is 4 more than the second.

Puzzle 2: A play sold 30 tickets for $171. Adult tickets cost $8, student tickets $5, and child tickets $3. There were twice as many student tickets as adult tickets.

Part 2 · Notes and examples (fill in each note, then work the example)

Solution: A solution of a three-variable system is an ordered _____.

Elimination: Use two equations to eliminate a variable, then a _____ pair to eliminate the _____ variable.

Example 1

Solve the system.

$x - 2y + z = -11$

$3x + 2y - z = 7$

$-x + 2y + 4z = -9$

Special cases: 0 = 5 means _____; 0 = 0 means _____.

Example 2

Solve the system.

$x + y + z = 2$

$5x + 5y + 5z = 3$

$4x + y - 3z = -6$

Example 3

Solve the system.

$x = 2z$

$y + z = 3$

$2x - y + z = 9$

Example 4

A furniture store advertises: sofa and love seat, $1,250; sofa and two chairs, $1,400; sofa, love seat, and one chair, $1,550. What is the price of each piece?

Part 3 · Partner practice

Partner 1

Solve the system.

$x + y + z = 6$

$x - y + z = 2$

$2x + y - z = 1$

Partner 2

The sum of three numbers is 12. The second is 3 times the first, and the third is 2 more than the first. Find the numbers.